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SHORT RESEARCH AND DISCUSSION ARTICLE

Metagenomic exploration reveals a differential patterning of antibiotic resistance genes in urban and peri-urban stretches of a riverine system

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Abstract

Antimicrobial resistance in the riverine ecosystem of urban areas is an alarming concern worldwide, indicating the importance of molecular monitoring to understand their patterning in urban and peri-urban areas. In the present study, we evaluated the influence of urban rivers on the connected peri-urban rivers of a riverine system of India in the context of antibiotic resistance genes. The rivers traversing through urban (Mula, Mutha, Pawana, and Ramnadi) and peri-urban stretches (Bhima and Indrayani) form the riverine system of Pune district in Maharashtra, India. The MinION-based shotgun metagenomic analysis revealed the resistome against 26 classes of antibiotics, including the last line of antibiotics. In total, we observed 278 ARG subtypes conferring resistance against multiple drugs (40%), bacitracin (10%), aminoglycoside (7.5%), tetracycline (7%), and glycopeptide (5%). Further, the alpha diversity analysis suggested relatively higher ARG diversity in the urban stretches than peri-urban stretches of the riverine system. The NMDS (non-metric multidimensional scaling) analysis revealed significant differences with overlapping similarities (stress value = 0.14, p-value = 0.004, ANOSIM statistic R: 0.2328). These similarities were reasoned by assessing the influence of downstream sites (sites at the outskirts of Pune city; however, directly impacted), which revealed significant differences in the ARG contents of urban and peri-urban stretches (stress value = 0.14, p-value = 0.001, ANOSIM statistic R: 0.6137). Overall, we detected the dissemination of antibiotic resistance genes from the polluted urban rivers into the peri-urban rivers located downstream in the connected riverine system potentially driven by anthropogenic activities.

Keywords MinION · Peri-urban rivers · Urbanization · Antibiotic resistance

Introduction

Contamination of the natural environment with active pharmaceutical ingredients (antibiotics) is rapidly increasing amid its unrestricted and widespread use. The release of contaminated wastewaters from various industries, hospitals, agriculture, animal production, and aquaculture into the aquatic bodies facilitates the emergence and evolution and potentially triggers the dissemination of antibiotic-resistant genes (ARGs) (Singh et al. 2019). Globally, India is one of the largest consumers of antibiotics and shows higher irrational usage of antibiotics (Marathe et al. 2017). Consequently, there appears to be a higher incidence of antibiotic resistance, which is reflected in the water bodies of India, resulting in their dismal conditions (Singh et al. 2019, Ahammad et al. 2014, Marathe et al. 2017, Yadav et al. 2021, Water Quality Database, Borthakur and Singh 2016). Several studies have revealed the enormity of ARGs in important and highly revered Indian rivers (Reddy and Dubey 2019, Diwan et al. 2017, Yadav et al. 2020, Samson et al. 2019). Reddy and Dubey, in 2019, reported 23 different ARG types in River Ganga, whereas Yadav et al. (2020) reported 29 different ARG types in the river Bhima and Indrayani (142 and 333 ARG subtypes, respectively).

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The peri-urban rivers possess hybrid urban–rural features and are irreversibly crucial as they provide water for various purposes, including drinking water and agriculture irrigation (Yadav et al. 2021; Zheng et al. 2018; Chen et al. 2019a, 2019b). Despite their significance, these rivers are subjected to uncontrolled wastewater discharges from various sources. Several reports worldwide have revealed diverse ARGs in these peri-urban river ecosystems (Zheng et al. 2018; Chen et al. 2019a, 2019b; Zhang et al. 2020).

Pune city is the ninth largest city in India, located in Maharashtra, with a population near 6 million. The rivers Mula, Mutha, Mula-Mutha (formed after the confluence of Mula and Mutha River), Ramnadi, Pawana, Indrayani, and Bhima are the major rivers flowing through the Pune district and form a unique riverine system. The rivers Mula, Mutha, Mula-Mutha, Ramnadi, and Pawana primarily traverse through the highly populated and polluted urban settlements of Pune city. The Ramnadi and Pawana are the tributaries of Mula River, meeting at the right and left bank, respectively, before Mula River meets Mutha River at the heart of Pune city to form Mula-Mutha River (Fig. 1). Further, the Mula-Mutha River flows downstream (away from Pune city) and unloads its content into Bhima River (mostly flowing through peri-urban and agriculture dominant regions) at the outskirts of Pune city; eventually, Bhima River flows towards southern India and meets Krishna River, which empties itself in the Bay of Bengal.

In the last decade, the rivers of Pune have been reported for harboring diverse ARGs and ARBs, including the resistance to the last line of antibiotics (Marathe et al. 2017; Yewale et al. 2019; Razavi et al. 2017). However, these studies were limited to fewer stretches of the riverine system and did not address urban and peri-urban aspects. To the best of our knowledge, few efforts have systematically explored the ARGs of the interconnected urban and peri-urban rivers (Zheng et al. 2018; Chen et al. 2019a, 2019b; Zhang et al. 2020). River sediments harbor the maximum living biomass and are hugely responsible for the metabolic cycles (Gibbons et al. 2014). We recently elaborated microbial diversity differences in the urban and peri-urban stretches, reflecting the differential influence of urbanization and industrialization that accelerated anthropogenic activities (Yadav et al. 2021). Here, we hypothesized dissimilarity in the composition of ARGs in the interconnected urban and peri-urban stretch amid the differential level of urbanization. The MinION-based shotgun sequencing of river sediments was performed to examine the differences of ARGs in the urban and peri-urban stretch and understand the overall ARG structure in the riverine system.

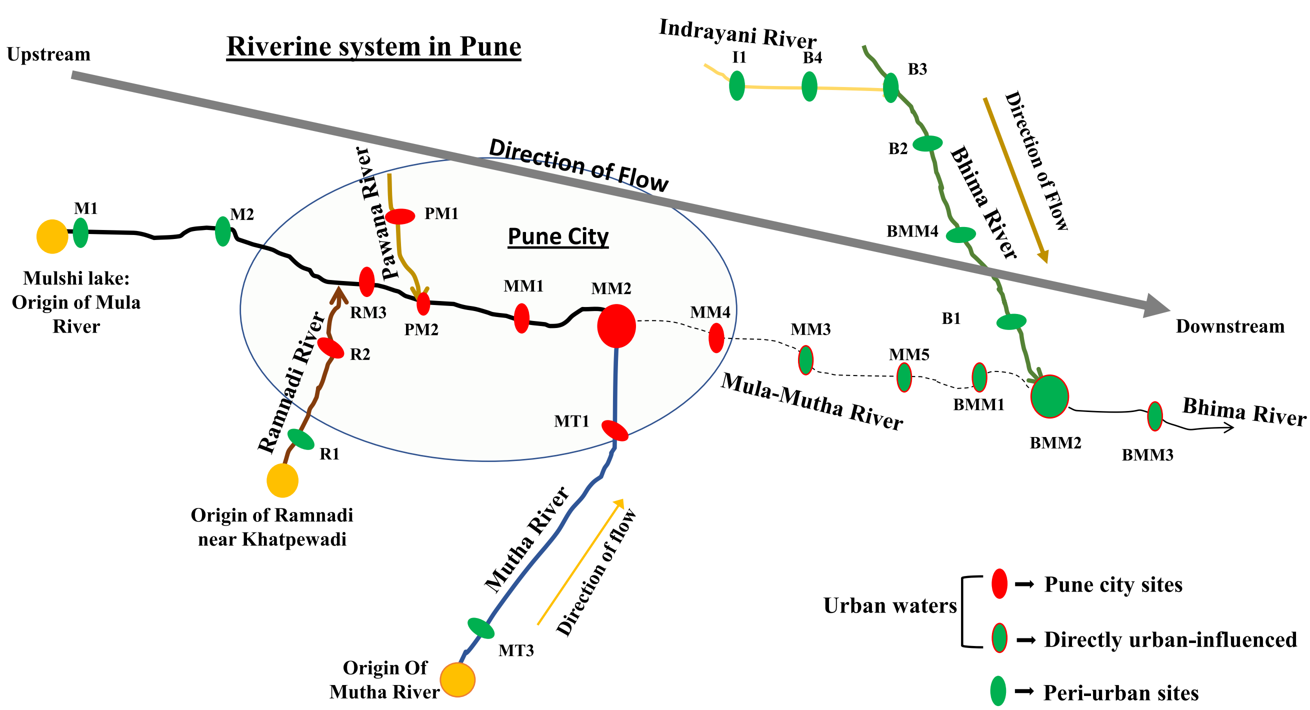

Fig. 1 Schematic of riverine system. The illustration displays the sampling sites and describes the groupings of sampling sites

Materials and methods

Sample site description and sample collection

Based on geographic region, population density, and previous studies, we divided the riverine system into urban waters (sites in the Pune city and downstream sites of Mula-Mutha River) and peri-urban waters (sites upstream of Mula River, sites of River Indrayani and River Bhima (except BMM3)) (Dhawde et al. 2018, Census India 2011, Hui and Wescoat Jr 2019) (Online Resource 1, 2 and Fig. 1). River sediment samples (10–15 cm from the top) were collected during December 2018 in sterile 250 ml containers and stored at 4 °C until further processing. Details of sampling sites and coordinates are given in Online Resource 1. Urban sites ($n = 13$) include MM3 MM4, MM5, BMM1, BMM2, BMM3, MT1, R2, RM3, PM1, PM2, MM1, and MM2, and peri-urban sites ($n = 10$) include M1, M2, R1, MT3, I1, B1, B2, B3, B4, and BMM4.

MinION library preparation and sequencing

Metagenomic DNA was extracted from 400 mg of river sediment using DNeasy PowerSoil Pro Kit (Qiagen). The quality and quantity of DNA were assessed by NanoDrop Lite Spectrophotometer (Thermo Fisher Scientific) and Qubit fluorometer (Thermo Fisher Scientific) using broad range (BR) high sensitivity (HS) assay kit (Thermo Fisher Scientific, Q32853 and Q32851, respectively). The library preparation was carried out using 1D ligation sequencing kit SQK-LSK109, and barcoding was done with native barcoding EXP-NBD104 and NBD114. Around 500 ng of the library was loaded on FLO-MIN 106D R9 Flow cells. The data is submitted in NCBI Sequence Read Archive (SRA) repository (Online Resource 1).

Data processing and analysis

The reads were basecalled using Albacore (v2.3.4), followed by demultiplexing and trimming with qcat (v1.0.1) and porechop (v0.2.4). The antibiotic resistance genes and mobile genetic elements were predicted using a long-read specific NanoARG tool (Arango-Argoty et al. 2019) with default parameters. NanoARG tool uses deep learning-based approach to predict ARGs. Further, stringency was put by selecting only those ARG reads, which showed a minimum bit score of 60 and identity of 30 (Pearson 2013). The alpha diversity and non-metric multidimensional scaling (NMDS) was constructed (using Bray–Curtis dissimilarity measure) using R packages phyloseq and microbiome (McMurdie and Homes 2013, Lahti and Shetty 2017).

Statistical analysis

The alpha diversity was performed using the unfiltered gene data, as we observed only 3X difference between the minimum and the maximum number of ARG reads per sample. The significance of alpha diversity was assessed using the Welch two-sample *t*-test. The variable sequence depth was adjusted using the DESEQ2 package (Love et al. 2014). ANOSIM (p-value threshold of 0.05, permutation of 999) was used to interpret the ordination analysis statistically. Beta-disper (p-value threshold of 0.05, permutation of 999) was used to assess the homogeneity. Fold change analysis was performed using METAGENassist (Arndt et al. 2012).

Results and discussion

Antibiotic resistance genes are now considered emerging environmental pollutants. They can freely disseminate and evolve independently of their host via various transfer routes, such as mobile genetic elements (Ouyang et al. 2015). Urbanization affects the movement of people in the urban surroundings, resulting in an increase in population density and industrialization to facilitate economic activities (Peng et al. 2020; Neiderud 2015). Various studies have implicated the correlation of urbanization with the increased prevalence of antibiotic resistance. Human activities such as irrational use of antibiotics, use of additives in animal and fish feedings, and release of partially or untreated wastewater are the multiple factors of urbanization that affect the riverine systems (Ouyang et al. 2015). These direct impacts of increasing anthropogenic activities on the urban rivers have a chain effect, polluting the connected downstream rivers in the outskirts of cities. To understand such a phenomenon concerning antibiotic resistance in the rivers, we used a metagenomic approach to study the differences in the resistome of a riverine system composed of urban and peri-urban stretches. Overall, the nanopore sequencing generated 3.64 million reads yielding around 18.9 Gbases with an average quality score and sequence length of 8.86 and 5.1 kb, respectively.

Occurrence and distribution of ARGs

The aquatic environment provides a suitable ecological niche for the emergence and dissemination of antibiotic-resistant microbes due to ARG transfer between autochthonous and allochthonous bacteria (Almakki et al. 2019). The analysis revealed ARGs against 26 categories of antibiotics, including the last resort antibiotics such as polymyxin in the riverine system. Overall, 40% of the ARGs were classified to *multidrug* resistance (MDR) genes, followed by 10% conferring resistance to *bacitracin*, 7.5% to *aminoglycoside*, 7% to *tetracycline*, and 5% to *glycopeptide* (Fig. 2). The other highly prevalent ARGs (0.1%) were detected against the categories such as *macrolides*, *lincosamides*, and *streptogramines* (MLS), *fosmidomycin*, *fluoroquinolone*, *beta-lactam*, *peptide*, *rifamycin*, *sulfonamide*, *phenicol*, *diaminopyrimidine*, *polymyxin*, *triclosan*, *aminocoumarin*, *nucleoside*, *pleuromutilin*, and *fosfomycin* (Fig. 2). The fold change analysis revealed a significant increase of ARGs against

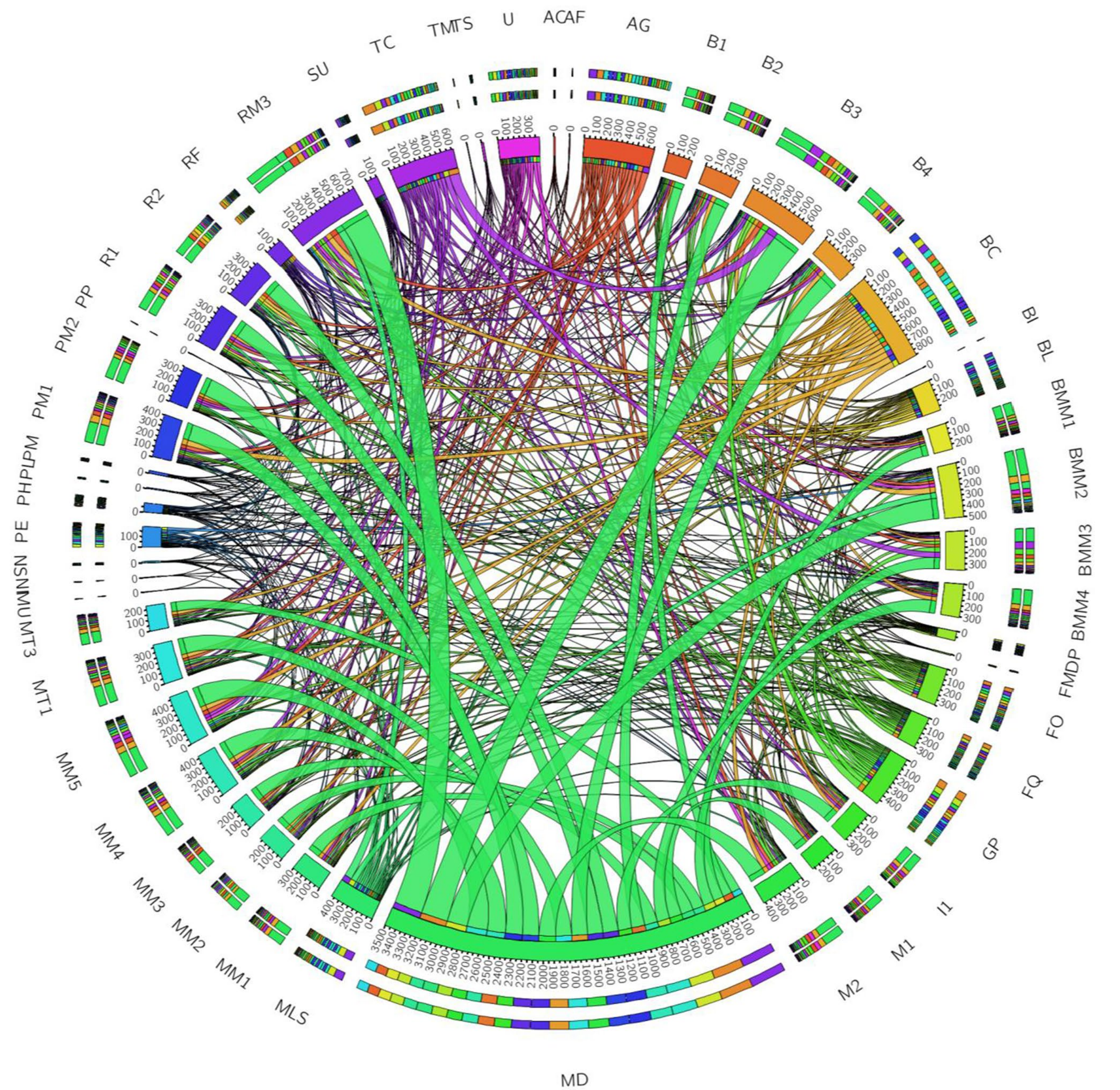

Fig. 2 Resistance against different categories of antibiotics. The circos plot illustrates the detection of different antibiotic resistance genes against 26 categories of antibiotics. GP, glycopeptide; BL, beta-lactam; PP, polyamine and polypeptide; MD, multidrug; SU, sulfonamide; AG, aminoglycoside; FQ, fluoroquinolone; BC, bacitracin; PE, peptide; TC, tetracycline; FO, fosmidomycin; U, unclassified; MLS, macrolides, lincosamides, and streptogramines; RF, rifamycin; PH, phenicol; DP, diaminopyrimidine; AC, aminocoumarin; PM, polymyxin; FM, fosfomycin; TM, tetracenomycin C; NS, nucleoside; PL, pleuromutilin; TS, triclosan; NI, nitroimidazole; M, mupirocin; BI, bicyclomycin; AF, antibacterial free fatty acids

fosfomycin, *nitroimidazole*, *sulfonamide* (*t*-test with FDR correction, $p < 0.05$), and *tetracenomycin C* in the urban and urban-influenced sites (Online Resource 3). At the same time, the peri-urban region showed enrichment of ARGs against polyamine and polypeptide group of antibiotics. In all, the total ARGs (against antibiotic classes) were more extensive in this riverine system as compared to the other polluted rivers of India, such as the River Ganges and River Yamuna (Das et al. 2020, Reddy and Dubey et al. 2018), and were comparable to urban and peri-urban rivers across the world (Chen et al. 2019c, 2019d; Zheng et al. 2018; Singh et al. 2019). Relatively, the ARGs against six categories, including *antibacterial free fatty acids*, *tetracenomycin*, *polyamine/polypeptide*, *nitroimidazole*, *mupiromycin*, and *bicyclomycin*, were scarcely identified (Fig. 2).

Altogether, the analysis revealed 278 different ARG subtypes in the riverine system. The previous study by Marathe et al. (2017) on Mutha River identified 175 ARGs, the majority of them getting mapped to city samples. The ARGs such as *ompR*, *rpoB2*, *bcrA*, *bacA*, *tetA(48)*, *kdpE*, *vanR*, *multidrug ABC transporter*, *mtrA*, *rosB*, *oprM*, and *patA* were among the most dominant ARG (≥ 2%) subtypes in the riverine system (Fig. 3). The *ompR*, a regulator gene involved in significant upregulation of numerous MDR drug exporter genes (Hirakawa et al. 2003), was the most dominant ARGs, followed by *rpoB2* RNA polymerase variant, which evades the action of rifamycin (CARD database). Further, most of the detected ARGs such as *mtr*, *acr*, *ade*, *cme*, *mds*, *mdt*, *mex*, and *sme* (Online Resource 4) encoding resistance to multiple drugs provide antibiotic efflux as the central mechanism of resistance (CARD ontology and CARD database). We observed numerous genes for various beta-lactamases that efficiently hydrolyze many third-generation antibiotics. Several members of *VEB beta-lactamases* (also known as Vietnamese extended-spectrum beta-lactamases) were observed, such as *VEB-1*, *VEB-2*, *VEB-3*, and *VEB-7* (Online Resource 4). These extended-spectrum beta-lactamases (ESBLs) are known to inactivate broad-spectrum *cephalosporins* and *aztreonam* (CARD database). The analysis also revealed several other important beta-lactamases, which include *tem* gene that confers resistance against *cephalosporin*, *OXA beta-lactamases* that are reported in pathogens such as *Acinetobacter baumannii* and *Enterobacter*, and *NmcR beta-lactamase* commonly found in *Enterobacter* species (Palzkill 2013, McArthur et al. 2013). Previously, the mobile

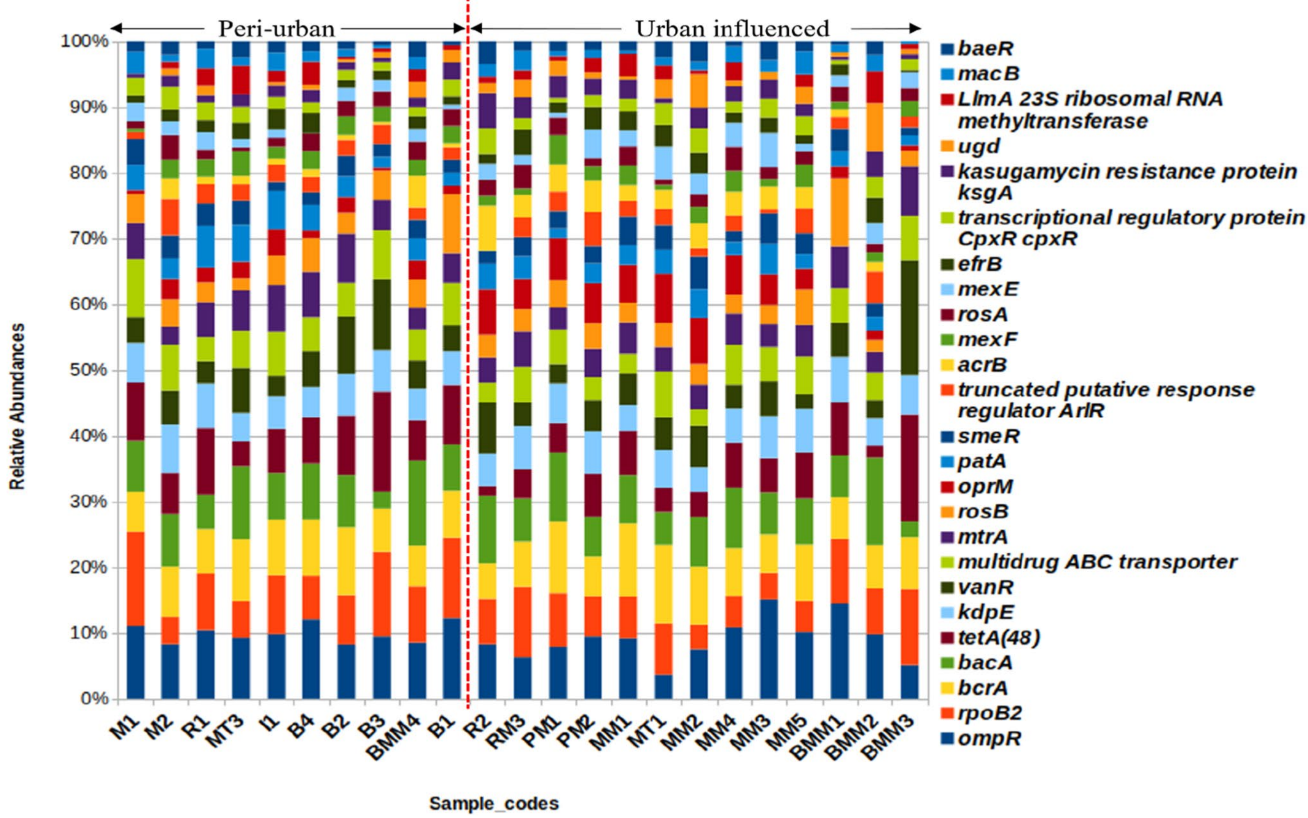

Fig. 3 Distribution of Antibiotic resistance genes. The relative abundance stacked bar plot illustrates the distribution of ARGs (≥ 1%) in the riverine system

colistin resistance gene *(mcr1)* conferring resistance to the last resort drug polymyxin was reported in Mutha River by Marathe et al. (2017). These *mcr* genes confer resistance against drugs such as colistin belonging to polymyxin, which is often considered the last line of antibiotics. The analysis revealed the polymyxin inactivating *mcr5* and *arnA* gene in the riverine system. Overall, the results indicated a higher prevalence of different ARG subtypes conferred against almost all major clinically relevant antibiotics.

Spatial diversity of ARGs in the riverine system and mobilome

Alpha and beta diversity analysis was performed to understand the overall differences in the ARGs of the urban and peri-urban stretch of the riverine system. We observed relatively higher alpha diversity (Shannon and Chao1 indices) of the ARGs in the urban stretch compared to the peri-urban stretch (Online Resource 5); however, it was not significant. The NMDS analysis, along with ANOSIM statistic, indicated significant diversity differences (p-value = 0.004, ANOSIM statistic R: 0.2328); however, the lower R value indicated the overlapping similarities (Fig. 4). This could probably be attributed to the similar climate conditions and the downstream sites MM3, MM5, BMM1, BMM2, and BMM3. We confirmed this by taking "rivers" as a variable. We observed no significant influence of rivers on the observed diversity differences of ARGs (p-value = 0.301, ANOSIM statistic R: 0.05679). Although geographically, the downstream sites MM3, MM5, BMM1, BMM2, and BMM3 are at the outskirts of Pune city, these sites are directly influenced by the urban waters, unlike the other peri-urban sites, and thus, these sites were grouped into urban waters. Further, the comparison of the urban waters (excluding the sites MM3, MM5, BMM1, BMM2, and BMM3) with the peri-urban sites revealed significant differences (p-value = 0.001) with higher ANOSIM statistic R of 0.6137. These observations strongly indicate the following. (i) These sites are potentially responsible for the overlapping similarities observed due to their dual characteristics of urban and peri-urban. (ii) We observed clear demarcation in the ARG diversity between the Pune city and peri-urban sites.

The mobile genetic elements (MGEs) (Online Resource 6) are responsible for disseminating ARGs, leading to an increase in multidrug resistance. We observed the dominance of transposases, integrases, and recombinases at all the sites. The MGEs such as insertion sequences (IS) and transposons are DNA segments carrying different ARGs that

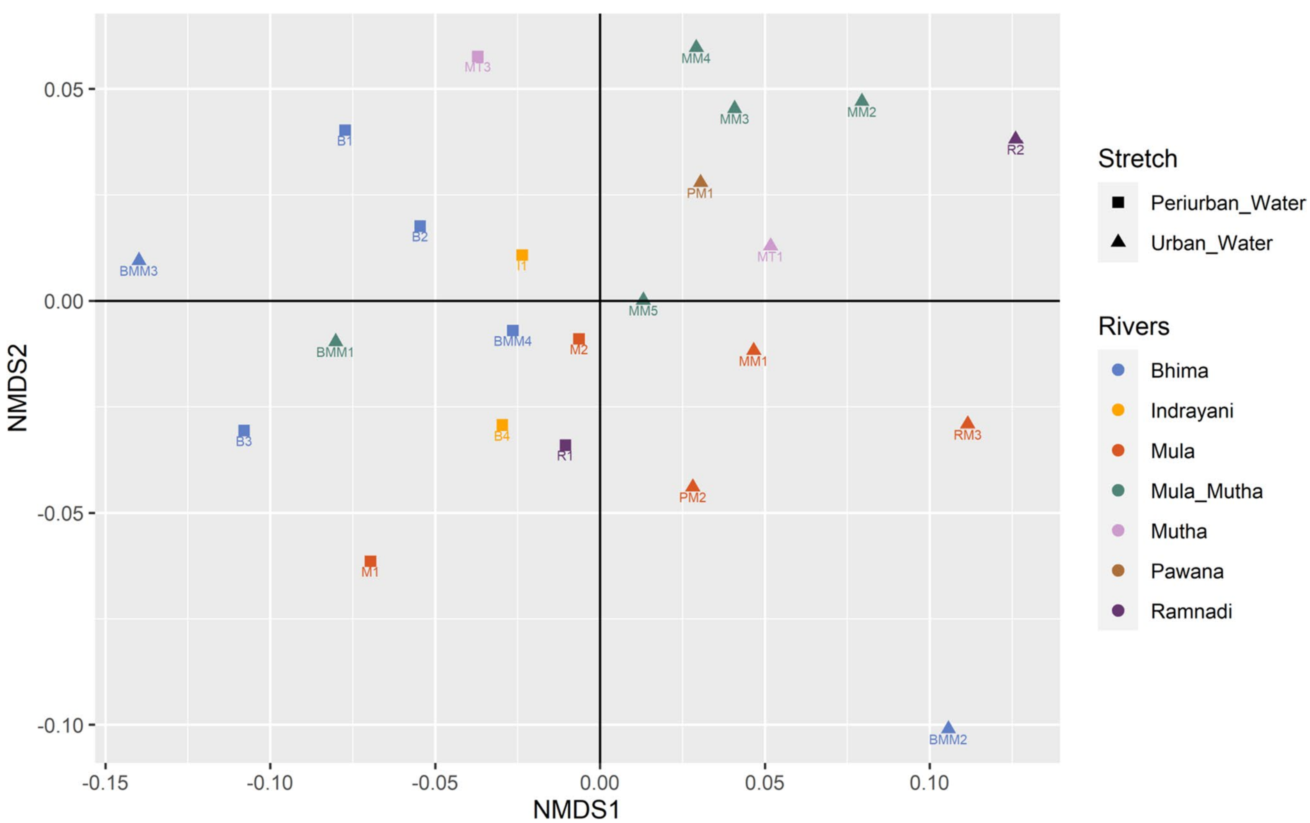

Fig. 4 Beta diversity of ARGs. The beta diversity was assessed using NMDS (stress value = 0.14) with the Bray–Curtis dissimilarity matrix. The statistical significance was estimated using ANOSIM (p-value = 0.004, ANOSIM statistic R: 0.2328)

can move themselves to random locations with the help of enzymatic machinery that involves transposases and recombinases. The other elements, such as integrons with the help of integrases and various recombinases, carry out site-specific recombination to spread resistance (Frost et al. 2005). The class 1 integron (intI1) detected in all sediment samples is one of the major MGEs present in various environmental microorganisms (Chen et al. 2020). Together these observations suggested the considerable impact of the urban waters on the connected downstream sites at the outskirts of city in the interconnected rivers.

Conclusion

In conclusion, we observed a relatively higher diversity of ARGs in the urban waters than the peri-urban stretches indicating the differential influence of urbanization on the stretches of a riverine system. Nonetheless, the NMDS analysis also suggested the impact of the urban waters on the connected downstream stretch implicating the cumulative impact of the urbanization on the riverine stretch. The analysis also reflected that the downstream peri-urban sites MM3, MM5, BMM1, BMM2, and BMM3 were directly impacted by the urban waters and thus exhibited overlapping similarities in the resistome. Further, we also observed ARGs against all the clinically important antibiotics, including the last line of antibiotics such as polymyxin. Overall, this study provides an overview of the ARGs prevalent in the riverine system and delineates the spatial differences of the antibiotic resistance genes. The results discussed in this study emphasize necessary strategies such as appropriate treatment of the wastewater, removal of active pharmaceutical ingredients from the wastewater, adequate planning of city drains, and prevention of unrestricted release of untreated wastewater to curb the inflow of urban resistome entering into the downstream peri-urban river stretches.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11356-021-16910-y.

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Author contribution All the authors have read and review the manuscript for submission. VR: Analysis and writing original draft. RKY: Editing, sample collection, and sequencing. MD: Supervision, designing study, and reviewing.

Availability of data and materials The data is submitted in NCBI Sequence Read Archive (SRA) repository (Refer to Online Resource 1).

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

Ahammad ZS, Sreekrishnan TR, Hands CL, Knapp CW, Graham DW (2014) Increased waterborne bla NDM-1 resistance gene abundances associated with seasonal human pilgrimages to the Upper Ganges River. Environ Sci Technol 48(5):3014–3020. https://doi.org/10.1021/es405348h

Almakki A, Jumas-Bilak E, Marchandin H, Licznar-Fajardo P (2019) Antibiotic resistance in urban runoff. Sci Total Environ. https://doi.org/10.1016/j.scitotenv.2019.02.183

Arango-Argoty GA, Dai D, Pruden A, Vikesland P, Heath LS, Zhang L (2019) NanoARG: a web service for detecting and contextualizing antimicrobial resistance genes from nanopore-derived metagenomes. Microbiome 7(1):1–18. https://doi.org/10.1186/s40168-019-0703-9

Arndt D, Xia J, Liu Y, Zhou Y, Guo AC, Cruz JA, Sinelnikov I, Budwill K, Nesbø CL, Wishart DS (2012) METAGENassist: a comprehensive web server for comparative metagenomics. Nucleic Acids Res 40(W1):W88–W95. https://doi.org/10.1093/nar/gks497

Borthakur A, Singh P (2016) India's lost rivers and rivulets. Energy Ecol Environ 1(5):310–314. https://doi.org/10.1007/s40974-016-0039-2

CARD database. https://card.mcmaster.ca/. Last accessed 06th May 2021

CARD ontology. https://card.mcmaster.ca/ontology/36005. Last accessed 06th May 2021

Census India 2011. https://www.censusindia2011.com/maharashtra/pune/pune-city-population.html. Last Accessed on 25th Mar 2021

Chen H, Li Y, Sun W, Song L, Zuo R, Teng Y (2020) Characterization and source identification of antibiotic resistance genes in the sediments of an interconnected river-lake system. Environ Int 137:105538. https://doi.org/10.1016/j.envint.2020.105538

Chen H, Bai X, Jing L, Chen R, Teng Y (2019a) Characterization of antibiotic resistance genes in the sediments of an urban river revealed by comparative metagenomics analysis. Sci Total Environ 653:1513–1521. https://doi.org/10.1016/j.scitotenv.2018.11.052

Chen H, Bai X, Li Y, Jing L, Chen R, Teng Y (2019b) Characterization and source-tracking of antibiotic resistomes in the sediments of a peri-urban river. Sci Total Environ 679:88–96. https://doi.org/10.1016/j.scitotenv.2019.05.063

Chen H, Bai X, Li Y, Jing L, Chen R, Teng Y (2019c) Source identification of antibiotic resistance genes in a peri-urban river using novel crAssphage marker genes and metagenomic signatures. Water Res 167:115098. https://doi.org/10.1016/j.envint.2020.105538

Chen H, Chen R, Jing L, Bai X, Teng Y (2019d) A metagenomic analysis framework for characterization of antibiotic resistomes in river environment: application to an urban river in Beijing. Environ Pollut 245:398–407. https://doi.org/10.1016/j.envpol.2018.11.024

Das BK, Behera BK, Chakraborty HJ, Paria P, Gangopadhyay A, Rout AK, Nayak KK, Parida PK, Rai A (2020) Metagenomic study focusing on antibiotic resistance genes from the sediments of River Yamuna. Gene 758:144951. https://doi.org/10.1016/j.gene.2020.144951

Dhawde R, Macaden R, Saranath D, Nilgiriwala K, Ghadge A, Birdi T (2018) Antibiotic resistance characterization of environmental E. coli isolated from River Mula-Mutha, Pune District, India. Int J Environ Res Publ Health 15(6):1247. https://doi.org/10.3390/ijerph15061247

Diwan V, Purohit M, Chandran S, Parashar V, Shah H, Mahadik VK, Stålsby Lundborg C, Tamhankar AJ (2017) A three-year follow-up study of antibiotic and metal residues, antibiotic resistance and resistance genes, focusing on Kshipra—a river associated with holy religious mass-bathing in India: protocol paper. Int J Environ Res Public Health 14(6):574. https://doi.org/10.3390/ijerph14060574

Frost LS, Leplae R, Summers AO, Toussaint A (2005) Mobile genetic elements: the agents of open source evolution. Nat Rev Microbiol 3(9):722–732

Gibbons SM, Jones E, Bearquiver A, Blackwolf F, Roundstone W, Scott N, Hooker J, Madsen R, Coleman ML, Gilbert JA (2014) Human and environmental impacts on river sediment microbial communities. PLoS ONE 9(5):e97435. https://doi.org/10.1371/journal.pone.0097435

Hirakawa H, Nishino K, Hirata T, Yamaguchi A (2003) Comprehensive studies of drug resistance mediated by overexpression of response regulators of two-component signal transduction systems in Escherichia coli. J Bacteriol 185(6):1851–1856. https://doi.org/10.1128/JB.185.6.1851-1856.2003

Hui R, Wescoat JL Jr (2019) Visualizing peri-urban and rurban water conditions in Pune district, Maharashtra, India. Geoforum 102:255–266. https://doi.org/10.1016/j.geoforum.2018.01.008

Lahti L, Shetty S (2017) Tools for microbiome analysis in R. Microbiome package

Love MI, Huber W, Anders S (2014) Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biol 15:550. https://doi.org/10.1186/s13059-014-0550-8

Marathe NP, Pal C, Gaikwad SS, Jonsson V, Kristiansson E, Larsson DJ (2017) Untreated urban waste contaminates Indian river sediments with resistance genes to last resort antibiotics. Water Res 124:388–397. https://doi.org/10.1016/j.watres.2017.07.060

McArthur AG, Waglechner N, Nizam F, Yan A, Azad MA, Baylay AJ, Bhullar K, Canova MJ, De Pascale G, Ejim L, Kalan L (2013) The comprehensive antibiotic resistance database. Antimicrob Agents Chemother 57(7):3348–3357. https://doi.org/10.1128/AAC.00419-13

McMurdie PJ, Holmes S (2013) phyloseq: an R package for reproducible interactive analysis and graphics of microbiome census data. PloS one 8(4). https://doi.org/10.1371/journal.pone.0061217.

Neiderud, C. J. (2015). How urbanization affects the epidemiology of emerging infectious diseases. Infection ecology & epidemiology, 5(1), 27060. https://doi.org/10.3402/iee.v5.27060

Ouyang WY, Huang FY, Zhao Y, Li H, Su JQ (2015) Increased levels of antibiotic resistance in urban stream of Jiulongjiang River China. Appl Microbiol Biotechnol 99(13):5697–5707. https://doi.org/10.1007/s00253-015-6416-5

Palzkill T (2013) Metallo-β-lactamase structure and function. Ann Ny Acad Sci 1277:9. https://doi.org/10.1111/j.1749-6632.2012.06796.x

Peng F, Guo Y, Isabwe A, Chen H, Wang Y, Zhang Y, Zhu Z, Yang J (2020) Urbanization drives riverine bacterial antibiotic resistome more than taxonomic community at watershed scale. Environ Int 137:105524. https://doi.org/10.1016/j.envint.2020.105524

Pearson WR (2013) An introduction to sequence similarity ("homology") searching. Curr Protoc Bioinformatics 42(1):3–1. https://doi.org/10.1002/0471250953.bi0301s42

Razavi M, Marathe NP, Gillings MR, Flach CF, Kristiansson E, Larsson DJ (2017) Discovery of the fourth mobile sulfonamide resistance gene. Microbiome 5(1):160. https://doi.org/10.1186/s40168-017-0379-y

Reddy B, Dubey SK (2019) River Ganges water as reservoir of microbes with antibiotic and metal ion resistance genes: high throughput metagenomic approach. Environ Pollut 246:443–451. https://doi.org/10.1016/j.envpol.2018.12.022

Samson R, Shah M, Yadav R, Sarode P, Rajput V, Dastager SG, Dharne MS, Khairnar K (2019) Metagenomic insights to understand transient influence of Yamuna River on taxonomic and functional aspects of bacterial and archaeal communities of River Ganges. Sci Total Environ 674:288–299. https://doi.org/10.1016/j.scitotenv.2019.04.166

Singh R, Singh AP, Kumar S, Giri BS, Kim KH (2019) Antibiotic resistance in major rivers in the world: a systematic review on occurrence, emergence, and management strategies. J Clean Prod 234:1484–1505. https://doi.org/10.1016/j.jclepro.2019.06.243

Water Quality Database, http://www.cpcbenvis.nic.in/water_quality_data.html, last accessed on 25th Feb 2021

Yadav R, Rajput V, Dharne M (2021) Metagenomic analysis of a mega-city river network reveals microbial compositional heterogeneity among urban and peri-urban river stretch. Science of The Total Environment, 146960. https://doi.org/10.1016/j.scitotenv.2021.146960

Yadav R, Rajput V, Gohil K, Khairnar K, Dharne M (2020) Comprehensive metagenomic insights into a unique mass gathering and bathing event reveals transient influence on a riverine ecosystem. Ecotoxicol Environ Saf 202:110938. https://doi.org/10.1016/j.ecoenv.2020.110938

Yewale PP, Lokhande KB, Sridhar A, Vaishnav M, Khan FA, Mandal A, Swamy KV, Jass J, Nawani N (2019) Molecular profiling of multidrug-resistant river water isolates: insights into resistance mechanism and potential inhibitors. Environ Sci Pollut 1-14. https://doi.org/10.1007/s11356-019-05738-2

Zhang Y, Chen H, Jing L, Teng Y (2020) Ecotoxicological risk assessment and source apportionment of antibiotics in the waters and sediments of a peri-urban river. Sci Total Environ 139128. https://doi.org/10.1016/j.scitotenv.2020.139128

Zheng J, Zhou Z, Wei Y, Chen T, Feng W, Chen H (2018) High-throughput profiling of seasonal variations of antibiotic resistance gene transport in a peri-urban river. Environ Int 114:87–94. https://doi.org/10.1016/j.envint.2018.02.039